

Optimal Control

Task 1M. Optimal control with constrained input

FengJiacheng 345276

Var 6

6	$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = u \end{cases}$	$\begin{aligned} x_1(0) &= x_2(0) = 0, \\ x_1(\pi) &= 2, \\ x_2(\pi/2) &= 0 \end{aligned}$
---	---	--

System Description

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = u$$

$$x_1(0) = x_2(0) = 0$$

$$x_1(\pi) = 2$$

$$x_2(\pi/2) = 0$$

Minimum-Time Optimal Control

Performance Index

$$J = t_f = \int_0^{t_f} dt$$

Hamiltonian

$$H = 1 + \lambda_1 x_2 + \lambda_2 u$$

Costate Equations

$$\dot{\lambda}_1 = 0$$

$$\dot{\lambda}_2 = -\lambda_1$$

$$\lambda_1 = c_1, \quad \lambda_2 = -c_1 t + c_2$$

Optimal Control Law

Hamiltonian is linear to u so:

$$u^*(t) = -\text{sign}(\lambda_2)$$

$$u^*(t) = \begin{cases} 1, & 0 \leq t < 10 \\ -1, & 10 \leq t \leq 20 \end{cases}$$

State Trajectories

$$x_2 = t, \quad x_1 = \frac{1}{2}t^2$$

$$x_2 = 20 - t$$

$$x_1 = 100 - \frac{1}{2}(20 - t)^2$$

So tf = 20

Minimum-Fuel Optimal Control

Hamiltonian

$$H = |u| + \lambda_1 x_2 + \lambda_2 u$$

Optimal Control Law

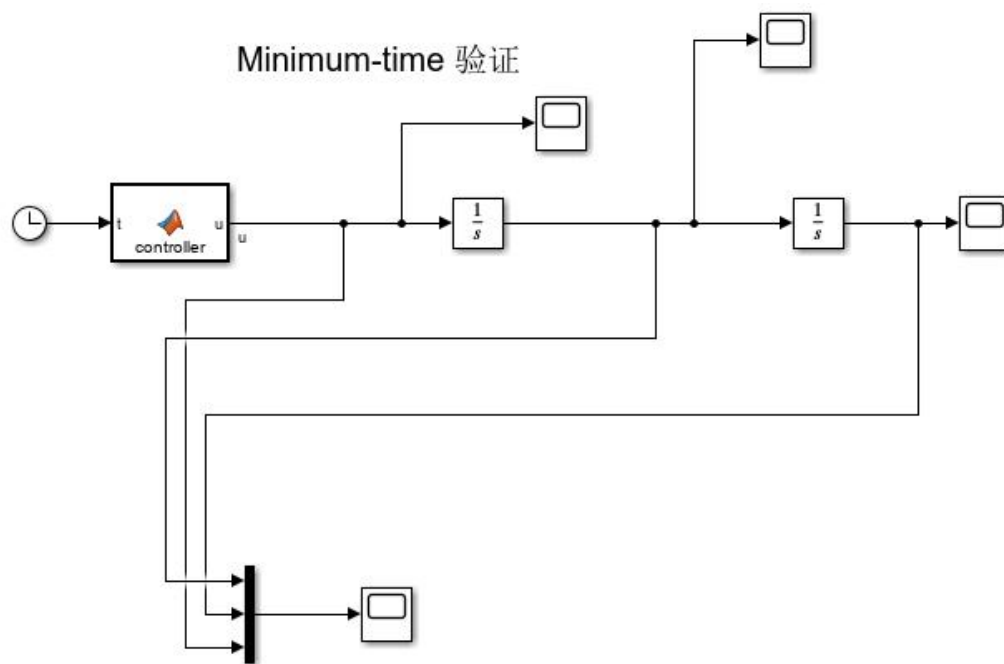
$$u^*(t) = \begin{cases} 1, & \lambda_2 < -1 \\ 0, & |\lambda_2| < 1 \\ -1, & \lambda_2 > 1 \end{cases}$$

$$a = \frac{T - \sqrt{T^2 - 400}}{2}$$

$$T = 30$$

$$a \text{ almost} = 3.82$$

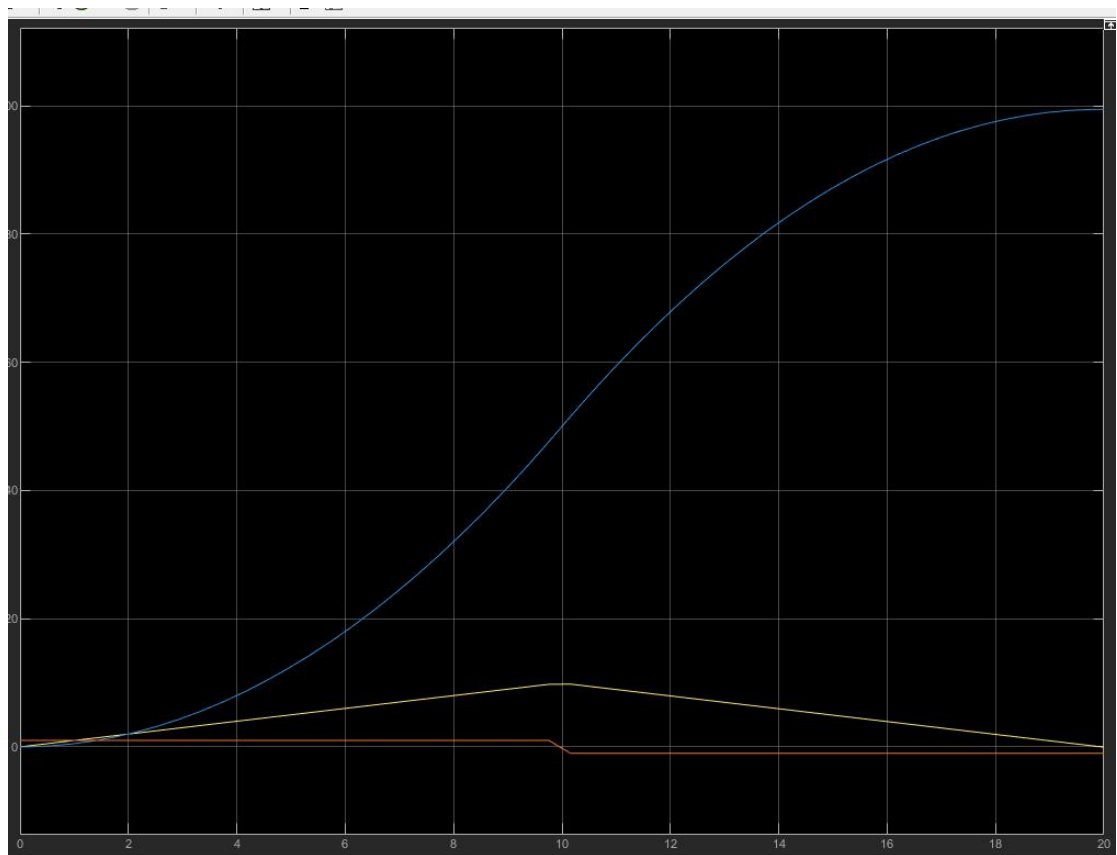
Simulink Verification



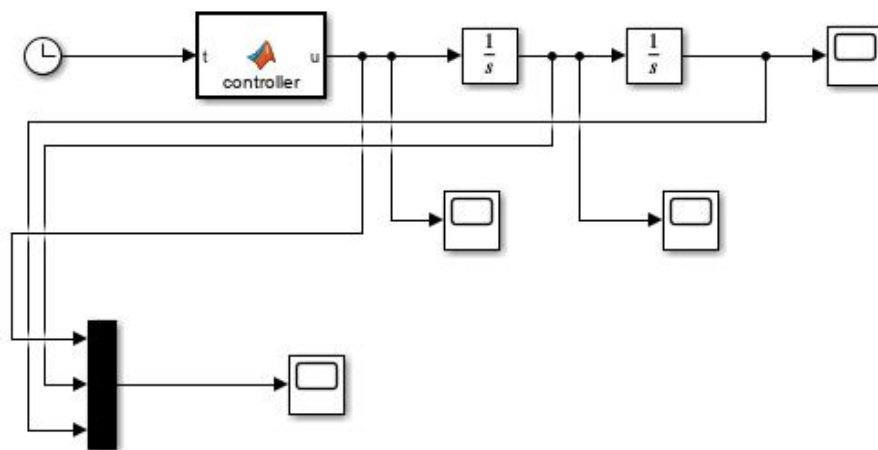
MATLAB FUNCTION

```
function u = controller(t)

if t < 10
    u = 1;
else
    u = -1;
end
```



Fuel-optimal 验证 $T = 30$ change the matlab function



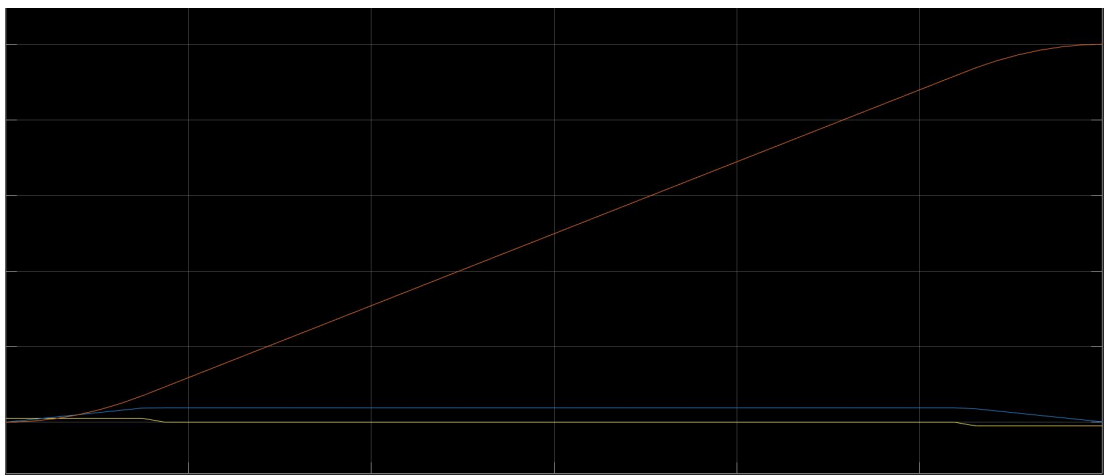
```

function u = controller(t)

T = 30; % 注意这里要和仿真终止时间一致
a = (T - sqrt(T^2 - 400))/2;

if t < a
    u = 1;
elseif t < T-a
    u = 0;
else
    u = -1;
end

```



Conclusion:

In this report, minimum-time and minimum-fuel control of a double integrator system were studied.

The minimum-time solution used bang-bang control, while the minimum-fuel solution used bang-off-bang control.

The Simulink results matched the theoretical analysis and satisfied the final conditions.

